

— Rout Dutta 1040683 AME → HW 3

RTS $\longrightarrow \frac{\partial}{\partial t} C(x,t) = \frac{1}{\lambda} \frac{\partial}{\partial x} \left(x \frac{\partial}{\partial x} C(x,t) \right) \dots (i)$
 (2) subject to $C(x,0) = 1$.

Let's start with separation of variables.

$$C(x,t) = X(x) T(t) \dots (ii)$$

$\therefore$ substituting (ii) in (i)
 $\Rightarrow X \cdot \frac{\partial}{\partial t} T = \frac{1}{\lambda} \frac{\partial}{\partial x} \left[x \frac{\partial}{\partial x} X \right]$

Dividing both sides by $X \cdot T$.

$$\frac{1}{T} \cdot \frac{\partial}{\partial t} T = \frac{1}{X} \frac{1}{\lambda} \frac{\partial}{\partial x} \left(x \frac{\partial}{\partial x} X \right)$$

Let us define a constant λ and use it as
 $\left\{ \begin{array}{l} \frac{1}{T} \frac{\partial}{\partial t} T = -\lambda^2 \quad \text{AND} \quad \frac{1}{X} \frac{1}{\lambda} \frac{\partial}{\partial x} \left(x \frac{\partial}{\partial x} X \right) = -\lambda^2 \end{array} \right\}$
 $\dots (iii) \quad \dots (iv)$

Solving (iii)

$$\frac{\partial}{\partial t} T = -\lambda^2 T \Rightarrow T(t) = \alpha e^{-\lambda^2 t} \dots (v)$$

Now $C(x,0) = X(x) \cdot T(0) = 1 \Rightarrow T(0) = 1$
 $\dots (vi)$

Supposing (vi) on (v)
 $T(0) = \alpha = 1$

Solving (iv)

$$\frac{1}{X} \frac{1}{\lambda} \frac{\partial}{\partial x} \left(x \frac{\partial}{\partial x} X \right) = -\lambda^2 X$$

Expanding the derivative we get

$$\frac{1}{X} \left(\frac{dX}{dx} + \frac{1}{x} \frac{dX}{dx} \right) + \lambda^2 X = 0$$

$$d_x \left(x \frac{d}{dx} X \right) + \lambda^2 X x = 0.$$

$$\Rightarrow \frac{d}{dx} X + x \frac{d^2}{dx^2} X + \lambda^2 X x = 0.$$

$$\Rightarrow \frac{d^2}{dx^2} X + \frac{1}{x} \frac{d}{dx} X + \lambda^2 X = 0.$$

$$\Rightarrow x^2 \frac{d^2}{dx^2} X + x \frac{d}{dx} X + \lambda^2 x^2 X = 0.$$

Bessel's equation of order 0 with $\arg = \{\lambda x\}$.

$$\therefore X(x) = A J_0(\lambda x) + B Y_0(\lambda x). \quad \begin{cases} J \rightarrow \text{Bessel 1st kind} \\ Y \rightarrow \text{Bessel 2nd kind} \end{cases}$$

Given constraint $\longrightarrow \left. \frac{\partial C}{\partial x} \right|_{x=1} = 0.$

$$\therefore \left. \frac{\partial X}{\partial x} \right|_{x=1} = -\lambda \left[A J_1(\lambda x) + B Y_1(\lambda x) \right] \Big|_{x=1} = 0.$$

$$\therefore A J_1(\lambda) = -B Y_1(\lambda)$$

Since A & B are normalization factors for our eigenfunctions, we can encode/take

$$A = Y_1(\lambda) \quad \& \quad B = -J_1(\lambda).$$

$$\phi_{\lambda}(x) = Y_1(\lambda) J_0(\lambda x) - J_1(\lambda) Y_0(\lambda x).$$

$\phi_{\lambda}(x)$ is our eigenfunction ansatz.

Given constraint $\longrightarrow \left. x \frac{\partial C}{\partial x} \right|_{x=\beta} = c(\beta, t).$
[BC 2].

$$\Rightarrow x (-\lambda) \left[Y_1(\lambda) J_1(\beta \lambda) - J_1(\lambda) Y_1(\beta \lambda) \right]$$

$$= Y_1(\lambda) J_0(\beta \lambda) - J_1(\lambda) Y_0(\beta \lambda).$$

rearranging these terms we get.

$$Y_1(\lambda) [Y_0(\lambda\beta) + \alpha\lambda Y_1(\lambda\beta)] = Y_1(\lambda) [Y_0(\lambda\beta) + \alpha\lambda Y_1(\lambda\beta)]$$

This is the eigenvalue equation that FORCES only ADMISSIBLE values of λ . ' λ ' is our eigenvalue. They are NOT arbitrary constants.

$$\begin{aligned} \text{B.C.3} \\ T(0) = 1 = \alpha \\ \alpha = 1 \\ \therefore T(x) = e^{-\lambda^2 x} \end{aligned}$$

$$\begin{aligned} \therefore C(x,t) &= \sum_n A_n \phi_n(x) \cdot e^{-\lambda_n^2 t} \\ &= \sum_{\lambda_n} A_n \phi_{\lambda_n}(x) \cdot e^{-\lambda_n^2 t} \end{aligned}$$

B.C.3

$$T(0) = 1.$$

$$\Rightarrow C(x,0) = \sum_{\lambda_n} A_n \phi_{\lambda_n}(x) = 1. \quad \text{--- (vi)}$$

Now, eigen functions are ORTHOGONAL

$$\therefore \langle \phi_{\lambda_n} | \phi_{\lambda_m} \rangle = \delta_{mn} \langle \phi | \phi \rangle.$$

Basically means, $\langle \phi_{\lambda_n} | \phi_{\lambda_m} \rangle$ is nonzero ONLY when $(m=n)$.

$\therefore$ Using (vi)

$$\sum_{\lambda_n} A_n \phi_{\lambda_n}(x) = 1.$$

Multiplying $\phi_{\lambda_k}(x)$ on both sides.

$$\sum_k A_k \left\{ \phi_{\lambda_k}(x) \cdot \phi_{\lambda_k}(x) \right\} = x \phi_{\lambda_k}(x)$$

Due to orthogonality $[K=1]$ ~~from~~ constraint boundaries at $x=1$ & $x=\beta$.

$$\therefore A_n \cdot x \phi_{\lambda_k}^2(x) = x \phi_{\lambda_k}(x)$$

$$\Rightarrow A_n = \frac{\int_{\beta}^1 x \phi_{\lambda_k}(x) dx}{\int_{\beta}^1 x \phi_{\lambda_k}^2(x) dx}$$

X

Assignment/HW - 3

Question 1 $\longrightarrow$ Rang Yusan.

Question 2 $\longrightarrow$ Rouni Dutta.